

Class 7 Lab

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2026-04-21

```
# Setup
# Load libraries
library(tidyverse) # includes ggplot2, tidyr, tibble, dplyr

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(pheatmap)
```

1. PCA of UK Food Data

The UK foods dataset contains weekly food consumption values for 17 food categories measured across four countries: England, Wales, Scotland, and Northern Ireland.

Data Import

```
url <- "https://tinyurl.com/UK-foods"
x <- read.csv(url)
```

Q1. How many rows and columns are in x? What functions can be used to find this?

```
dim(x)
```

```
[1] 17  5
```

```
nrow(x)
```

```
[1] 17
```

```
ncol(x)
```

```
[1] 5
```

The imported data frame has 17 rows and 5 columns. Functions that can be used to determine this include `dim()`, `nrow()`, and `ncol()`.

Checking data

```
head(x)
```

	X	England	Wales	Scotland	N.Ireland
1	Cheese	105	103	103	66
2	Carcass_meat	245	227	242	267
3	Other_meat	685	803	750	586
4	Fish	147	160	122	93
5	Fats_and_oils	193	235	184	209
6	Sugars	156	175	147	139

```
tail(x)
```

	X	England	Wales	Scotland	N.Ireland
12	Fresh_fruit	1102	1137	957	674
13	Cereals	1472	1582	1462	1494
14	Beverages	57	73	53	47
15	Soft_drinks	1374	1256	1572	1506
16	Alcoholic_drinks	375	475	458	135
17	Confectionery	54	64	62	41

The first column contains food names, but those names should really be row names. That is why the dataset has 5 columns instead of the expected 4 country columns.

```
rownames(x) <- x[,1]
x <- x[,-1]
head(x)
```

	England	Wales	Scotland	N.Ireland
Cheese	105	103	103	66
Carcass_meat	245	227	242	267
Other_meat	685	803	750	586
Fish	147	160	122	93
Fats_and_oils	193	235	184	209
Sugars	156	175	147	139

```
dim(x)
```

```
[1] 17 4
```

After fixing the row names, the dataset now has 17 rows and 4 columns, which matches the four UK countries.

Q2. Which row-name fix approach do I prefer and why?

I prefer re-reading the file with “row.names = 1”, as it is cleaner and easier to remember. The manual approach works, but if I accidentally run it multiple times, it will keep removing the first column and damage the dataset. Re-importing with “row.names = 1”, it avoids that problem and is safer for reproducible analysis.

```
x <- read.csv(url, row.names = 1)
head(x)
```

	England	Wales	Scotland	N.Ireland
Cheese	105	103	103	66
Carcass_meat	245	227	242	267
Other_meat	685	803	750	586
Fish	147	160	122	93
Fats_and_oils	193	235	184	209
Sugars	156	175	147	139

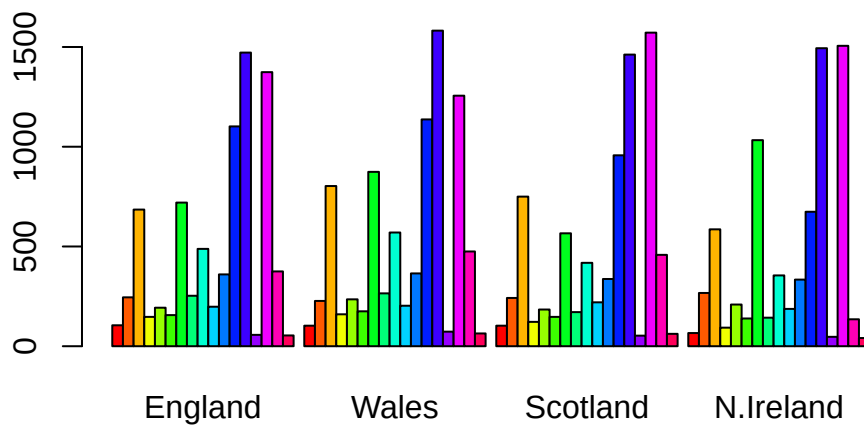
```
dim(x)
```

```
[1] 17  4
```

Spotting major differences and trends

Base R barplot

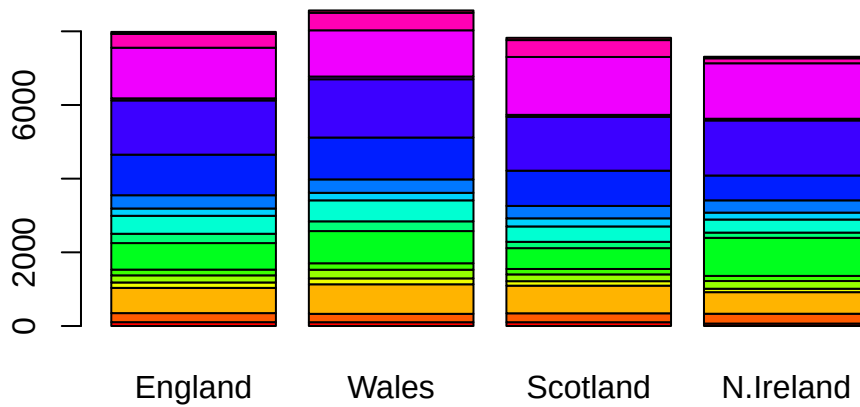
```
barplot(as.matrix(x), beside = TRUE, col = rainbow(nrow(x)))
```



Q3. What argument change creates a stacked version?

If I change “beside=TRUE” to “beside=FALSE” the bars become stacked instead of side-by-side. Leaving beside out has the same effect because the default is FALSE.

```
barplot(as.matrix(x), beside = FALSE, col = rainbow(nrow(x)))
```



Optional ggplot version with tidy data

In order to use ggplot2, the data needs to be converted from a wide format to long format.

```
x_long <- x |>
  rownames_to_column("Food") |>
  pivot_longer(
    cols = -Food,
    names_to = "Country",
    values_to = "Consumption"
  )

dim(x_long)
```

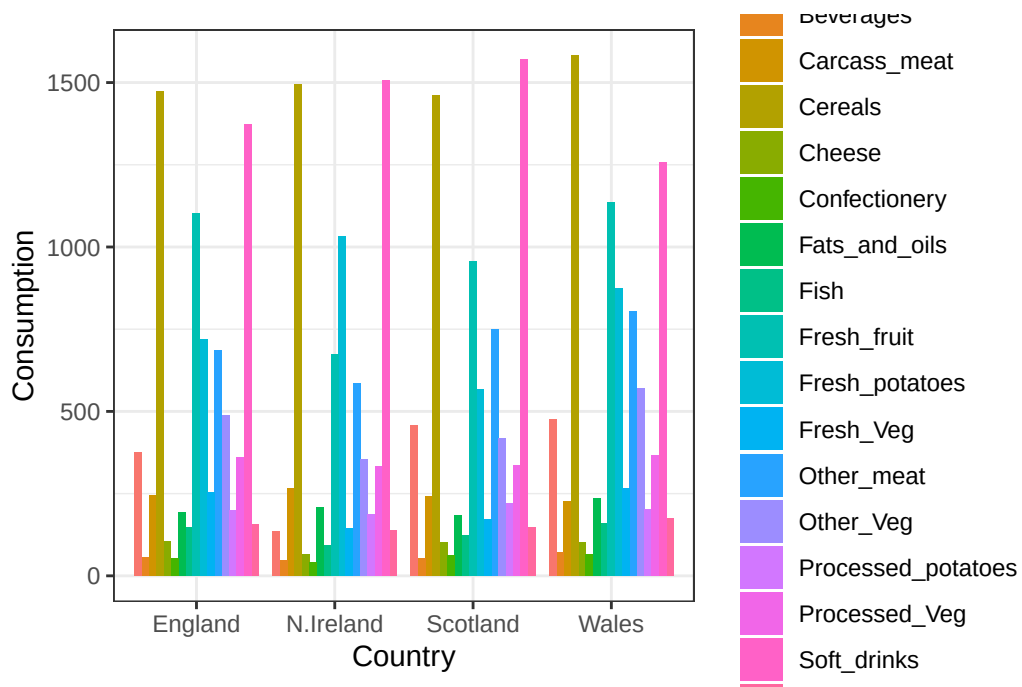
```
[1] 68  3
```

```
head(x_long)
```

```
# A tibble: 6 x 3
  Food      Country Consumption
<chr>      <chr>      <int>
```

1	"Cheese"	England	105
2	"Cheese"	Wales	103
3	"Cheese"	Scotland	103
4	"Cheese"	N.Ireland	66
5	"Carcass_meat "	England	245
6	"Carcass_meat "	Wales	227

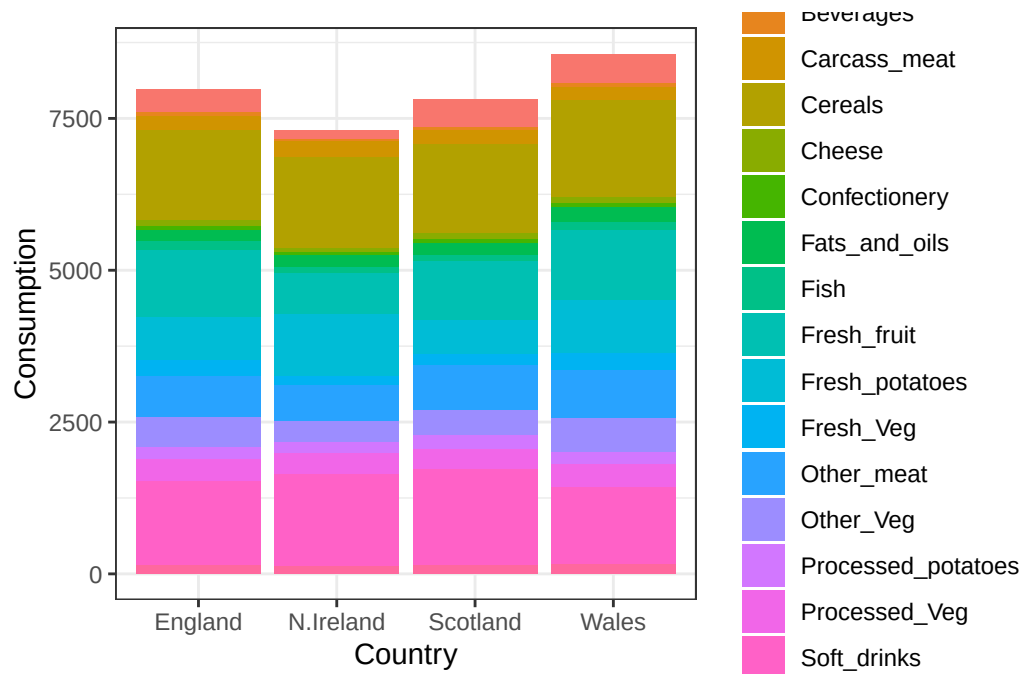
```
ggplot(x_long) +
  aes(Country, Consumption, fill = Food) +
  geom_col(position = "dodge") +
  theme_bw()
```



Q4. What argument change gives the stacked ggplot version?

To make a stacked barplot in ggplot 2, I can change the position from dodge to stack.

```
ggplot(x_long) +
  aes(x = Country, y = Consumption, fill = Food) +
  geom_col(position = "stack") +
  theme_bw()
```

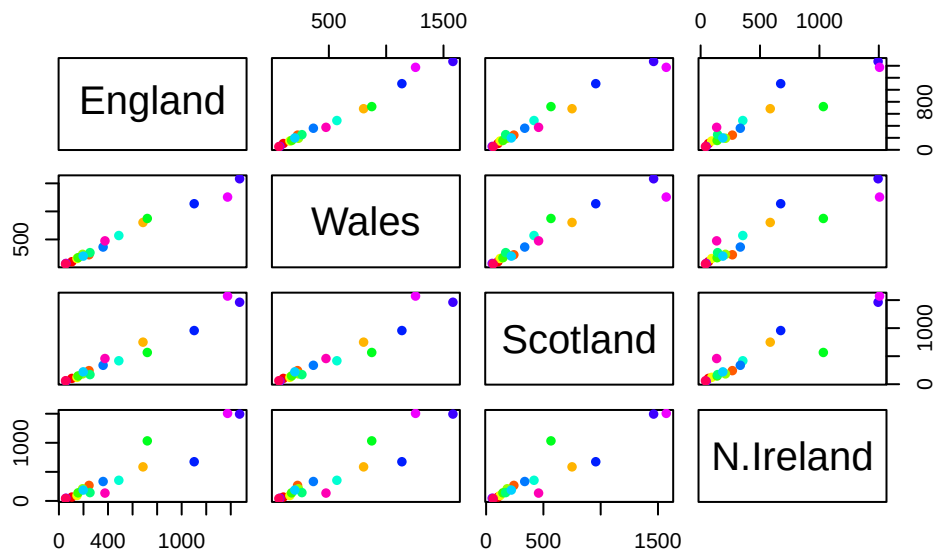


Pairs Plots and Heatmaps

Q5.

Pairplot

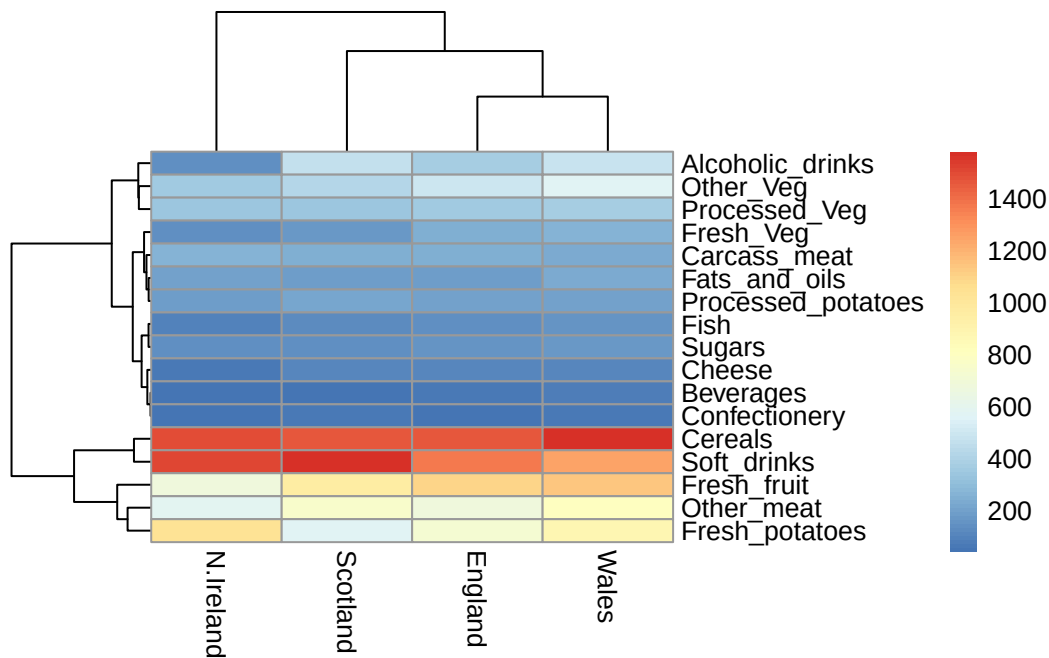
```
pairs(x, col = rainbow(nrow(x)), pch = 16)
```



This command makes all pairwise scatterplots between the four country variables. Each point represents one food category. If a point lies on the diagonal in a given pairwise comparison, that means the two countries have the same value for that food in that specific plot.

Heatmap

```
pheatmap(as.matrix(x))
```

Q6. Which countries cluster together, and what does this suggest?

From both the pairs plot and the heatmap, England, Wales, and Scotland cluster more closely together, while Northern Ireland appears more distinct. This suggests that England, Wales, and Scotland have more similar overall food consumption patterns, whereas Northern Ireland differs more strongly across several food categories.

It is not especially easy to identify the exact major differences just by looking at the raw pairs plot and heatmap alone, but Northern Ireland seems to stand out in foods such as fresh potatoes and soft drinks, while some of the other countries are associated more with foods like fresh fruit and alcoholic drinks.

PCA to the Rescue

The lab notes that `prcomp()` expects observations to be rows and variables to be columns, so the food table must be transposed before PCA. The page also reports that the first two PCs explain about 67.44% and 29.05% of the variance, respectively.

```
pca <- prcomp(t(x))
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	324.1502	212.7478	73.87622	2.921e-14
Proportion of Variance	0.6744	0.2905	0.03503	0.000e+00
Cumulative Proportion	0.6744	0.9650	1.00000	1.000e+00

Q7. Plot PC1 vs PC2

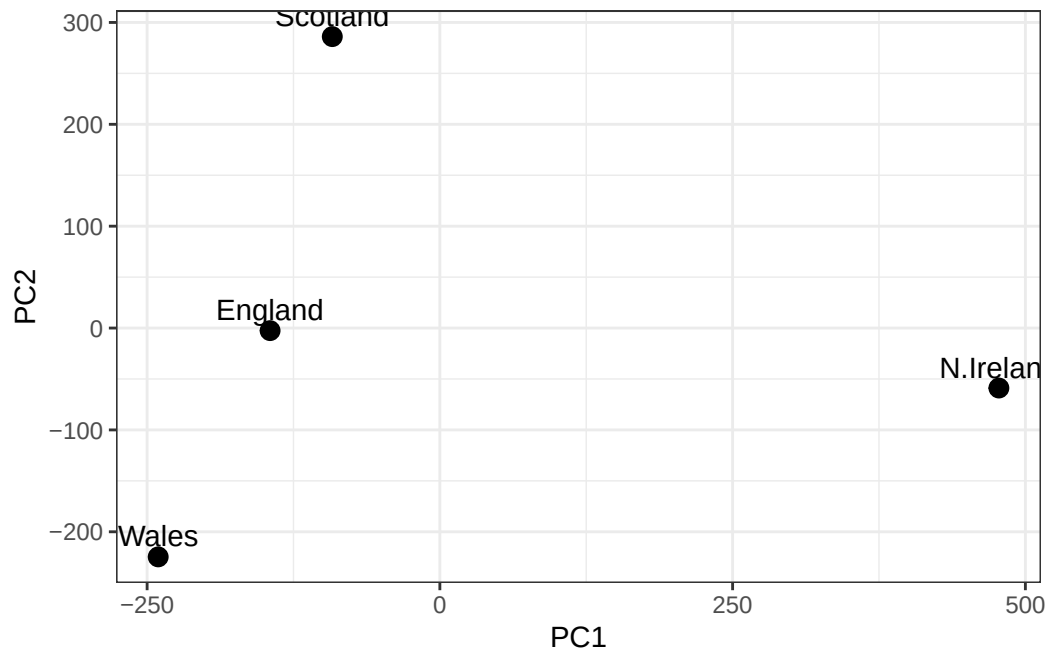
```
pca <- prcomp(t(x))
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	324.1502	212.7478	73.87622	2.921e-14
Proportion of Variance	0.6744	0.2905	0.03503	0.000e+00
Cumulative Proportion	0.6744	0.9650	1.00000	1.000e+00

```
df <- as.data.frame(pca$x)
df$Country <- rownames(df)

ggplot(df) +
  aes(PC1, PC2, label = Country) +
  geom_point(size = 3) +
  geom_text(vjust = -0.5) +
  theme_bw()
```

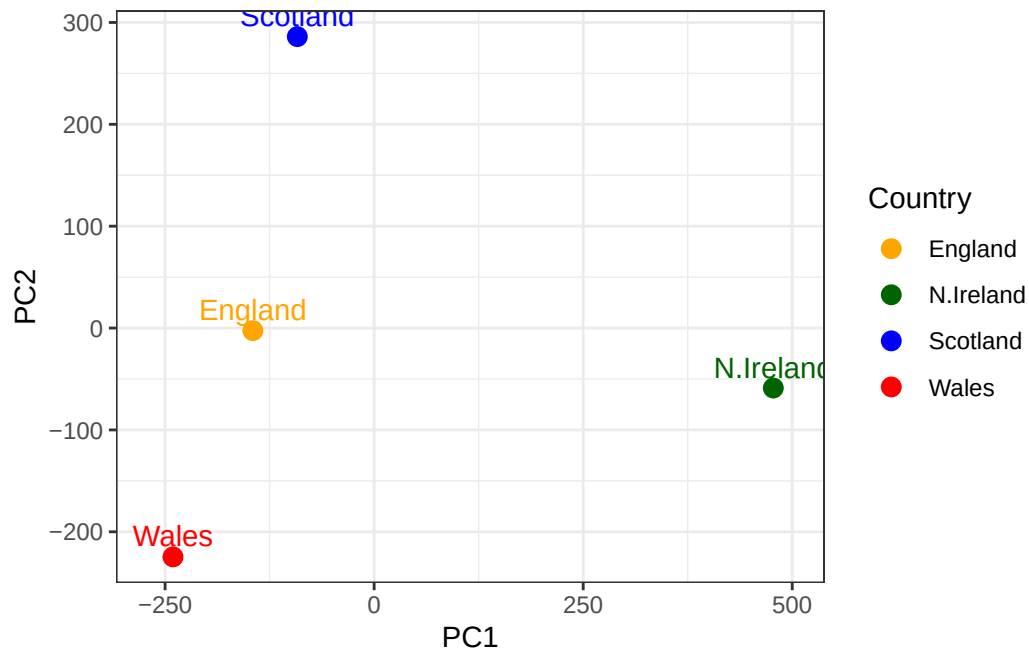


This PCA plot shows that England, Wales, and Scotland group relatively close together, while Northern Ireland is clearly separated, mainly along PC1.

Q8. Customize country label colors

```
country_cols <- c(
  "England" = "orange",
  "Wales" = "red",
  "Scotland" = "blue",
  "N.Ireland" = "darkgreen"
)

ggplot(df) +
  aes(x = PC1, y = PC2, label = Country, color = Country) +
  geom_point(size = 3) +
  geom_text(vjust = -0.5, show.legend = FALSE) +
  scale_color_manual(values = country_cols) +
  xlim(-270, 500) +
  xlab("PC1") +
  ylab("PC2") +
  theme_bw()
```



Variance explained

```
v <- round(pca$sdev^2 / sum(pca$sdev^2) * 100)
v
```

```
[1] 67 29 4 0
```

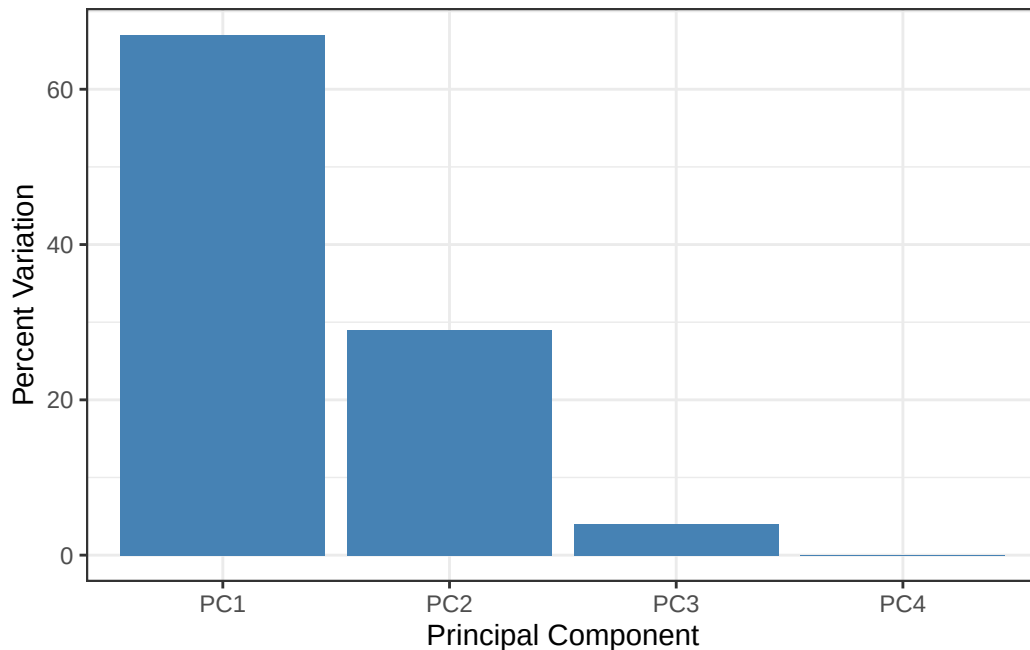
```
z <- summary(pca)
z$importance
```

	PC1	PC2	PC3	PC4
Standard deviation	324.15019	212.74780	73.87622	2.921348e-14
Proportion of Variance	0.67444	0.29052	0.03503	0.000000e+00
Cumulative Proportion	0.67444	0.96497	1.00000	1.000000e+00

The first principal component explains most of the variation, and the first two PCs together explain nearly all of it. This means that a two-dimensional PCA plot captures the major trends in the original 17-dimensional dataset very well.

Scree plot

```
variance_df <- data.frame(  
  PC = factor(paste0("PC", 1:length(v)), levels = paste0("PC", 1:length(v))),  
  Variance = v  
)  
  
ggplot(variance_df) +  
  aes(x = PC, y = Variance) +  
  geom_col(fill = "steelblue") +  
  xlab("Principal Component") +  
  ylab("Percent Variation") +  
  theme_bw()
```

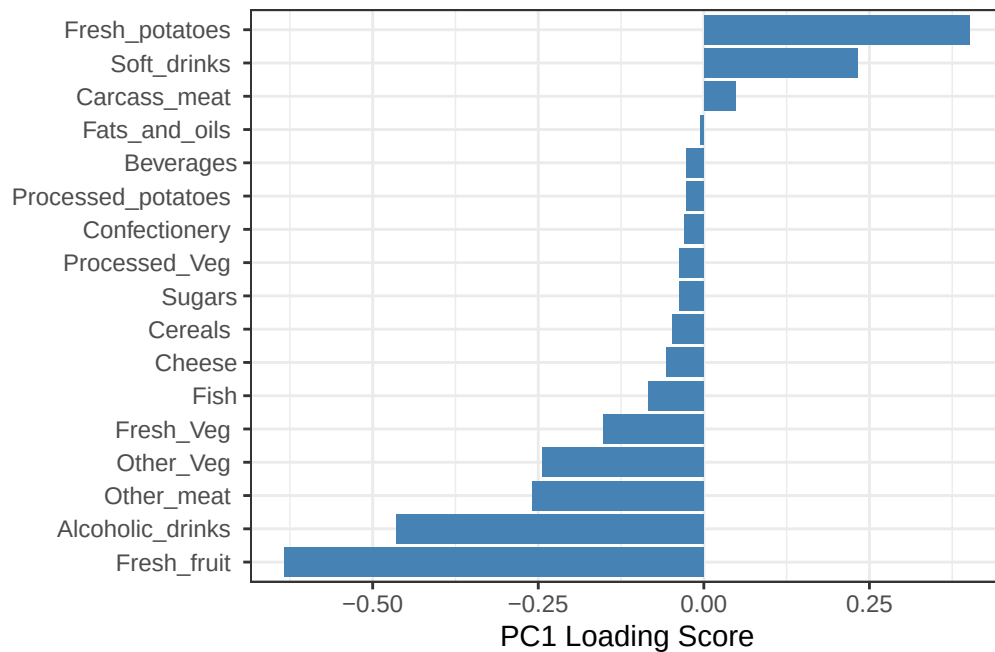


The scree plot confirms that **PC1 and PC2 dominate**, while the later components contribute very little additional variance.

Digging deeper: variable loadings

PC1 loading

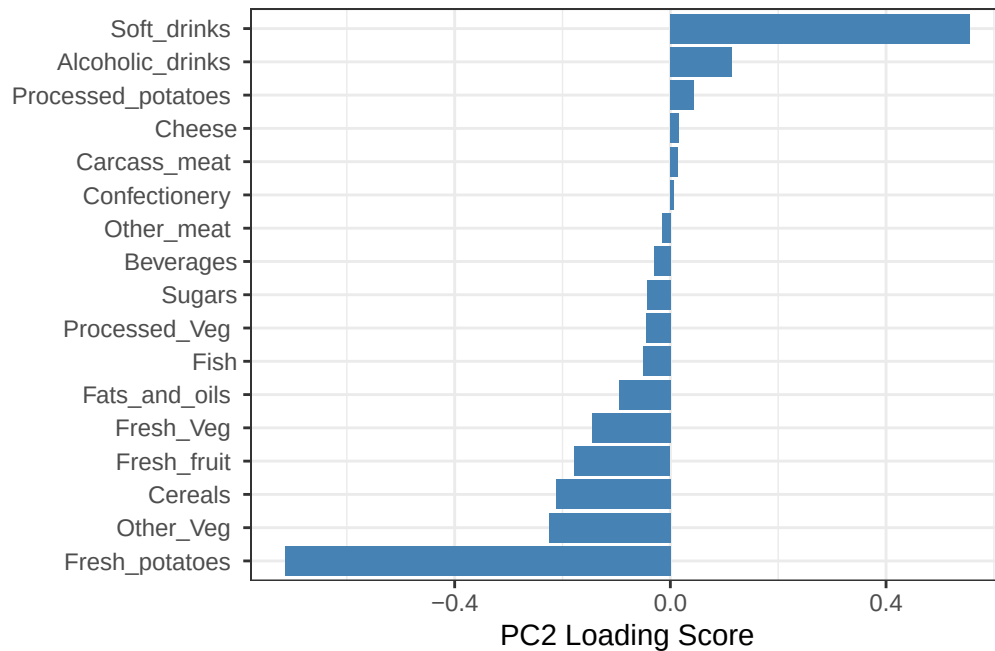
```
ggplot(as.data.frame(pca$rotation)) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col(fill = "steelblue") +
  xlab("PC1 Loading Score") +
  ylab("") +
  theme_bw() +
  theme(axis.text.y = element_text(size = 9))
```



The foods with the strongest positive PC1 loadings help push Northern Ireland to the positive side of the PCA plot, while foods with strong negative PC1 loadings help separate England, Wales, and Scotland to the other side.

Q9. PC2 loadings

```
ggplot(as.data.frame(pca$rotation)) +
  aes(x = PC2, y = reorder(rownames(pca$rotation), PC2)) +
  geom_col(fill = "steelblue") +
  xlab("PC2 Loading Score") +
  ylab("") +
  theme_bw() +
  theme(axis.text.y = element_text(size = 9))
```



PC2 appears to capture a secondary difference among the three more similar countries. The two most prominent food groups should be identified by inspecting the largest positive and negative bars on the PC2 loading plot. In general, PC2 tells us about the second-largest source of dietary variation after the strong Northern Ireland vs. rest-of-UK split captured by PC1.

Conclusion

Overall, PCA reduced a 17-dimensional food dataset to just two principal components that still preserve the vast majority of the variation. England, Wales, and Scotland cluster together, while Northern Ireland is more distinct. This makes PCA much more useful than trying to interpret the raw table or many pairwise plots alone.